Homework 3

Due Date: November 3, 2026 at 3pm

This homework will use the dataset "MaxART_HW3_2026.RData" in the Canvas homework folder.

When using AI to solve these problems, please be sure to include an appendix with your prompts, output and your method(s) of verification. The method of verification should be shown in sufficient detail so that we can evaluate its validity.

Question 1: Cluster-level analysis

This question's outcome variable is ``Y_bin'', where 1 means CD4 counts at 18 months were $\geq$ 350 cells/$mm^3$and 0 otherwise and ``clinc'' is the id for each clinic in this cluster-randomized trial.

a) MaxART is a SWD, so for the purposes of this exercise, we need to transform the treatment indicator to create a new 'Xcrt' variable, such that each clinic will be treated as a pure intervention or a pure control cluster. First, find the proportion of the treated participants in each clinic, then let Xcrt_k =1 if the proportion of the treated participants ranked in the top 7 clinics, and let Xcrt_k =0 if the proportion of the treated participants ranked in the bottom 7 clinics. Please create a concise well-labeled table that shows the proportion of treated participants for each cluster, and the cluster's treatment assignment.

b) Conduct a cluster-level analysis using the estimated outcome rates in each cluster, $\hat{p}_{ik} = \frac{\sum_{j=1}^{m_{ik}} Y_bin_{ijk}}{m_{ik}}$, and $\sigma_{ik}^2 = \frac{\hat{p}_{ik}(1-\hat{p}_{ik})}{m_{ik}}[1 + \rho(m_{ik} - 1)]$, where $i = 0, 1$ represents the control and treatment clusters. Calculate the cluster-level test statistic, p-value, estimated risk difference and risk ratio, and their 95% confidence intervals.

c) Compare the results of the cluster-level analysis with the individual-level results of the GEE model fit in R. For the risk difference, use the identity link function. For the risk ratio, use the log link function. Produce the model-based and empirical variances for these estimates. Provide a comparison of the individual-level results to the cluster-level results in a well-organized and labeled table.

d) Briefly interpret your findings. If there are differences in the two approaches, please concisely explain why.

Question 2: Joint optimization of (K, m).

a) Find the optimal K and m to minimize the cost of the parallel CRT created above with features otherwise similar to MaxART data, with at least 80% power. Let $c1$ = \$20,000 be the unit cost of adding a cluster, and $c2$ = \$100 be the unit cost of adding additional participants within a cluster. Find the optimal K and m such that the total cost of the study $Kc1 + Kmc2$ is minimized given the required power of this study to be 80% with a 5% Type 1 error rate. To obtain possible parameter values to inform the design of this new study similar to MaxART, please fit a GEE for binary outcomes with the identity link, where the outcome variable is "Y_bin" where 1 means CD4 counts at 18 months were $\geqslant$ 350 cells/mm3 and 0 otherwise, "clinc" is the id for each clinic in this cluster-randomized trial, and "Xcrt" is the treatment indicator. Assume constant cluster size.

b) Now suppose the per-cluster cost is drastically reduced to $c1$ = 200. What is the optimal K and m now?

c) In the actual study, there were 14 clusters (facilities). Is the optimal cluster size for 80% power achievable -- why or why not? If yes, what is the optimal cluster size? If not, what is the maximum power achievable?

d) How does the design optimized over (K, m) compare from the point of view of cost and power to the actual study conducted?

e) (Extra credit) Calculate the coefficient of variation of cluster size in MaxART and incorporate this parameter into the questions a-d above, using the Manatunga method and the van Breukelen method. Add these results to a well-labeled table and discuss the findings concisely.

Question 3: Marginal vs. conditional models (aka GEE vs GLMM, subject-specific vs. population-averaged models)

a) Derive the marginal model for binomial data with the identity link function and normally distributed random effects following
$E(Y_{ijk}|X_{ijk}, b_{ik}) = \beta_0^{\#} + \beta^{*\#} X_{ijk} + b_{ik}$. The marginal model is thus

$$E\left[Y_{ijk}\middle|X_{ijk},\beta^{\#}\right]=\int_{-\infty}^{\infty}[\beta_0+\beta^{\#}X_{ijk}+b\ ]f(b,\sigma_B^2)db,$$

with $b\sim N(0,\sigma_B^2)$.

b) Does $\beta^{\#}=\beta^{*\#}$?

c) In the parallel CRT created in question 1 above with features otherwise similar to MaxART data, fit these two models in R and report the results for $\beta^{\#}$, including point estimate, confidence interval and p-value, using both GEE and GLMM. Models should be specified as similarly as possible – exchangeable/compound symmetry used for assumed correlation between participants within a facility. Interpret your findings in subject matters terms, and compare the results from the two approaches from the point of view of interpretability and efficiency.